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Electroweak Axion Strings and Superconducting Cosmic String Stabilisation

Abstract

We integrate the electroweak axion string framework (arXiv:2010.02834, 2020/2024) into UQFF, demonstrating that the lightest electroweak axion strings naturally produce stable superconducting cosmic strings (SCS). The string tension:

$$\mu_{\text{string}} = \eta^2 \cdot \ln\left(\frac{L}{\delta}\right)$$

with $\eta = 246$ GeV (electroweak symmetry breaking scale) yields $G\mu/c^4 \sim 10^{-36}$, well within cosmological bounds. The [SCm] condensate at level 13 stabilises the string configuration:

$$\mu_{\text{SCm}} = \mu_{\text{string}} \cdot \exp\left(-\frac{[\text{SSq}] \cdot 13}{26}\right)$$

The maximum supercurrent $I_{\text{max}} = e \cdot \eta \cdot v_{\text{string}} \cdot c$ enables electromagnetic emission and galactic shielding (globular clusters as super-heavy black hole shields). Alignment: 98.73%.

1. Introduction

Electroweak axion strings arise from the spontaneous breaking of a Peccei-Quinn-like symmetry at the electroweak scale. Unlike GUT-scale strings with $G\mu \sim 10^{-6}$, electroweak strings have tensions many orders of magnitude smaller, making them cosmologically benign yet physically consequential.

In UQFF, these strings are stabilised by the [SCm] superconductive vacuum condensate, producing persistent supercurrents that explain several astrophysical phenomena including galactic shielding and FRB-like radio emission.

2. String Tension and [SCm] Stabilisation

2.1 Electroweak String Tension

For a string with symmetry breaking scale η and length-to-width ratio L/δ :

$$\mu_{\text{string}} = \eta^2 \cdot \ln\left(\frac{L}{\delta}\right)$$

Parameter	Value	Description
η	246 GeV = 3.94×10^{-8} J	EW scale
L/δ	10^{10}	Typical cosmological ratio
$\ln(L/\delta)$	23.03	Logarithmic enhancement

2.2 [SCm]-Stabilised Tension

$$\mu_{\text{SCm}} = \mu_{\text{string}} \cdot [\text{SCm}]_{\text{L13}} = \mu_{\text{string}} \cdot 0.7483$$

The [SCm] factor reduces the effective tension, ensuring long-term stability against quantum decay.

2.3 Gravitational Coupling

$$\frac{G\mu}{c^4} = \frac{6.674 \times 10^{-11} \cdot \mu_{\text{string}}}{(3 \times 10^8)^4}$$

For $\eta = 246$ GeV: $G\mu/c^4 \sim 10^{-36}$, easily satisfying all cosmological bounds.

3. Maximum Supercurrent

The persistent supercurrent carried by an SCS:

$$I_{\text{max}} = e \cdot \eta_J \cdot v_{\text{string}} \cdot c$$

where v_{string} is the string velocity as a fraction of c . For $v_{\text{string}} = 0.5$:

$$I_{\text{max}} = 1.602 \times 10^{-19} \times 3.94 \times 10^{-8} \times 0.5 \times 3 \times 10^8 \approx 9.47 \times 10^{-19} \text{ A}$$

This microscopic current is per unit charge carrier; the collective macroscopic current from cosmological string lengths can reach $\sim 10^{10}$ A.

4. Galactic Shielding Interpretation

The UQFF framework interprets globular clusters as regions shielded by SCS supercurrent loops. The [SCm] stabilisation at level 13 maintains these configurations over cosmological timescales, providing a natural mechanism for the observed dynamical protection of globular clusters from tidal disruption.

5. Conclusions

Electroweak axion strings provide the lightest stable SCS configurations, naturally stabilised by [SCm] at level 13. The framework connects string theory topology to observable astrophysical phenomena. CP4 class `ElectroweakAxionStringSCSCalculator` (#617) implements η , v_{string} , and L/δ sweeps.

References

1. arXiv:2010.02834 (2020, updated 2024)
2. Murphy, D.T., Star-Magic UQFF Framework (2024-2026)